Advanced Programming for Scientists

Lecture 1

Programming Languages Spectrum I

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Course Repository Introduction

v1.1

Prof. Mateusz Bawaj

A.D. 1308 unipg
UNIVERSITÀ DEGLI STUDI DI PERUGIA

Learning Objectives

By the end of this lecture you will:

- Understand the spectrum of scientific programming languages
- Motivate a language choice technically
- Understand why `git` is central to this course
- Navigate the course repository
- Open your first issue

Programming in Science ≠ Programming in Industry

Scientific software differs from generic software because:

- It often implements **mathematical models**
- It must be **reproducible**
- It frequently evolves with research questions
- It may scale from laptop experiments to HPC clusters
- **Often** written by non-professional programmers

Language Choice as Optimization Problem

Poor language choice can result in:

- Bottlenecks in simulation speed
- Inability to scale
- Difficult reproducibility
- High technical cost in long-term experiments

Language selection depends on:

- Performance constraints
- Development time
- Collaboration model

We can frame language choice as a model selection problem, similar to choosing a numerical solver or sampling strategy.

Why “Spectrum”?

There is no best programming language.

There is a language more suitable for:

- Type of problem
- Computational scale
- Performance constraints
- Collaboration model
- Ecosystem maturity

Language choice is an engineering decision, not an ideological one.

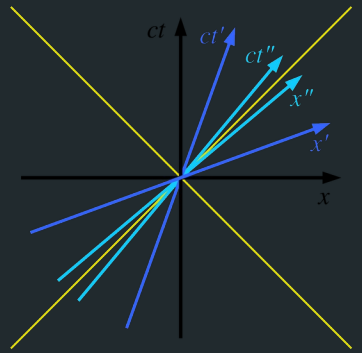
Lanuages integrate eachother. Heterogeneous approach may be the best one!

Learning programming language syntax is the last, not the first step of solving problems by software.

The programming language spectrum

Instead of ranking languages, we position them along axes:

- Execution speed
- Ecosystem maturity
- Ease of development
- HPC compatibility
- Long-term maintainability



Python
The scientific orchestrator

- Rapid prototyping
- Strong ecosystem
- Jupyter workflows
- ML integration
- Interfacing C/C++, etc.

Python
The scientific orchestrator

Strengths

- Extensive scientific ecosystem: NumPy, SciPy, matplotlib
- Rapid prototyping
- Integration with C/C++ and Fortran
- Strong ML ecosystem: TensorFlow, PyTorch, pandas, etc.
- Large community: stackoverflow
- Easy onboarding
- Portability

Python
The scientific orchestrator

Limitations

- Interpreter overhead
- Global Interpreter Lock (GIL)
- Hidden performance costs
- Requires hybridization for HPC

Python
The scientific orchestrator

Typical scientific uses

- Data analysis
- Signal processing
- Experiment control
- ML pipelines
- Prototyping

```python
print("Hello, World!")
```

C++ and Fortran
The numerical backbone

Used when:

- Solving Partial Differential Equations (PDEs)
- Large-scale Monte Carlo
- Fine memory control
- MPI / OpenMP

Performance as physical constraint

C++ and Fortran
The numerical backbone

Why They Still Matter

- Compiler optimization
- Deterministic performance
- Mature HPC ecosystem
- Legacy scientific codes

```fortran
program hello
  ! This is a comment line; it is ignored by the compiler
  print *, 'Hello, World!'
end program hello
```

```c
#include <stdio.h>

int main() {
  printf("Hello World!");
  return 0;
}
```

MATLAB
The laboratory ecosystem

- Signal processing
- Control systems
- Simulink workflows
- Rapid visualization

Discussion: reproducibility and licensing

Julia, R, Go
Other animals in the ZOO

Julia

- Just-In-Time (JIT) compiled
- Numerical focus

R

- Statistics dominant

Go

- Concurrency
- Systems tools

Risk vs innovation trade-off

Abstraction Layers in Scientific Software

Scientific code exists in layers:

- Mathematical model
- Numerical algorithm
- Programming language
- Compiler / Interpreter
- Machine code
- CPU instructions

All languages converge downward.

Abstraction Layers in Scientific Software

The CPU does NOT understand:

- Python, C++, Fortran, MATLAB, Julia

It executes:

- Machine instructions
- Binary opcodes
- Architecture-specific instruction sets

CPUs differ between themselves **A LOT**.

Abstraction Layers in Scientific Software

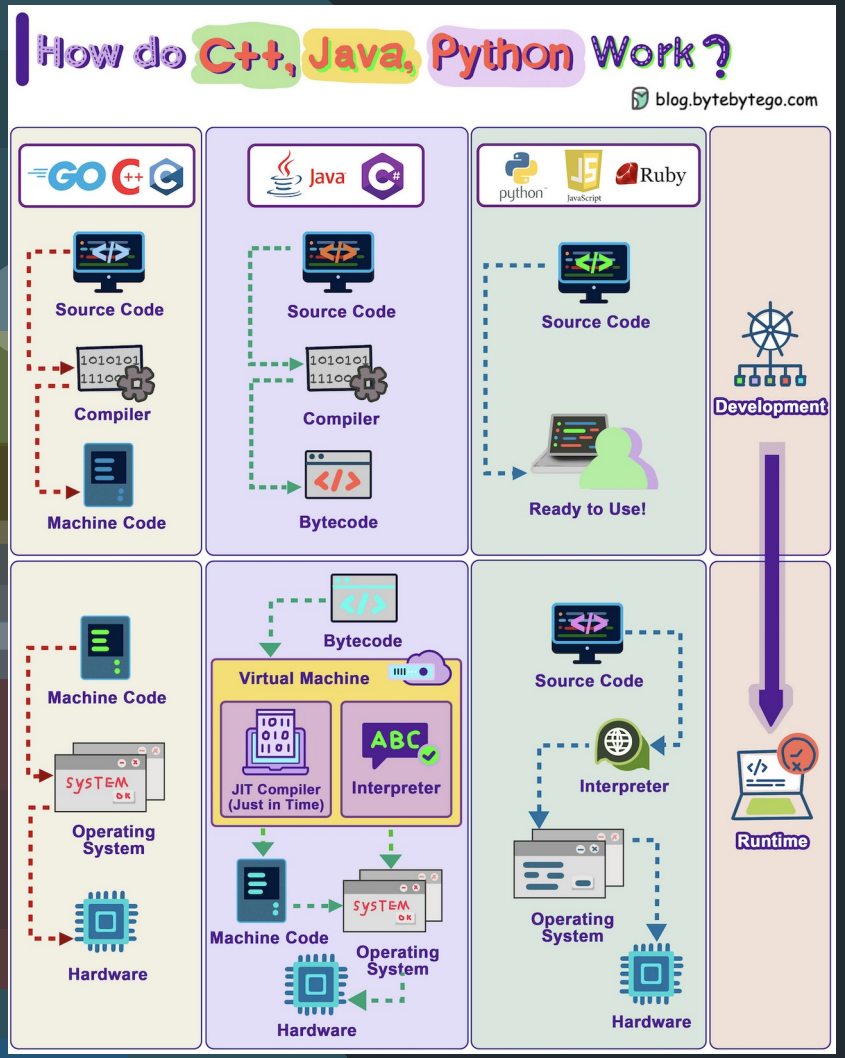
Everything becomes:

```
section .data
  hello:     db 'Hello world!',10    ; 'Hello world!' plus a linefeed character
  helloLen:  equ $-hello             ; Length of the 'Hello world!' string

section .text
  global _start

_start:
  mov eax,4            ; The system call for write (sys_write)
  mov ebx,1            ; File descriptor 1 - standard output
  mov ecx,hello        ; Put the offset of hello in ecx
  mov edx,helloLen     ; helloLen is a constant, so we don't need to say
                       ;  mov edx,[helloLen] to get it's actual value
  int 80h              ; Call the kernel
  mov eax,1            ; The system call for exit (sys_exit)
  mov ebx,0            ; Exit with return code of 0 (no error)
  int 80h;
```

Abstraction Layers in Scientific Software



Instruction Set Architecture (ISA)

Each CPU defines its own instruction set:

- x86_64
- ARM
- RISC-V

Scientific consequence:

- Binary compiled for one architecture
- Does not run on another

Hence:

- Need for recompilation
- Containerization
- CI testing across architectures

Hybrid Scientific Architecture

Common pattern:

- C++ computational core
- Python wrapper
- MPI scaling/Numbyfication
- Python analysis

Hybridization as design principle

Reproducibility Crisis in Scientific Code

Common problems:

- No version control
- No changelog
- No dependency tracking
- No peer review

Git

Git as Scientific Instrument

Git is:

- Versioned laboratory notebook
- Time machine
- Collaboration protocol
- Reproducibility layer

Not optional

Git
Course Git Workflow

Students will:

- Open Issues
- Discuss design
- Submit Pull Requests
- Review code
- Merge

Repository is living syllabus

Git

Issues as Scientific Questions

Issue examples:

- Performance bottleneck
- Refactor proposal
- Documentation improvement
- Language comparison

Issues = structured research problems

Git

Pull Requests as Peer Review

- Proposed change
- Transparent diff
- Discussion
- Approval
- Merge

Analogy: scientific journal review process

Git
Commit Messages as Lab Log

Bad commit:

- "fix stuff"

Good commit:

- "Replace nested loops with NumPy vectorization reducing runtime from 3.2s to 0.4s for $N=10^6$"

Precision matters, you comment for your future self!

In-Class Exercise

- Clone repository
- Explore structure
- Read CONTRIBUTING.md
- Open first Issue:
  - Describe your research software setup

Guided Reflection

For your research:

- Which language do you use?
- Why?
- Where are bottlenecks?
- Is it version controlled?
- What is missing?

Discussion

Would you redesign your software stack?

- Is your current language optimal?
- Could it benefit from Hybrid approach?
- It is HPC ready?

Key Takeaways

Language selection is:

- Scientific decision
- Performance decision
- Collaboration decision
- Sustainability decision

Git is:

- Infrastructure
- Quality control
- Collective memory

Homework

1)Prepare PC for work with git

2)Test the installation:

- Initialize or clean repository
- Add README and LICENSE
- Make three meaningful commits
- Open Issue describing architectural weakness